PRACTICAL -3

1.Aim:

Interface TM1637 with Raspberry Pi and Display current time.

2.Prior Concepts:

i) Concept of Python Programming

ii) Concept of 7 Segment Displays.

3.New Concepts:

i) Hardware connection of 7 seg with Raspi.

Display current time on 7 segments.

4. Hardware Requirement and PIN Connection.:

i) TM1637 4-digit seven segment Display board.

ii) Jumper wires

Pin Connection

TM1637 PIN	Function	RPi Pin	Function
1	GND(Ground)	6	Ground
2	VCC(+5V Power)	2	5V
3	DIN(Data In)	38	GPIO 20
4	CLK(Clock)	40	GPIO 21

5.Procedure:

Install Libraries of TM1637 as tm1637.py which is driver library. Download and save it in same folder as the main program code.

Write the below code:

```
#!/usr/bin/python

import time
import datetime
import tm1637 as obj

Display = obj. TM1637(CLK=21, DIO=20, brightness=5.0)
Display.Clear()

while(True):
        now = datetime.datetime.now()
        hour = now.hour
        minute = now.minute
        second = now.second
        Display.Clear()
        val = [(int(hour /10)), (hour % 10), (int(minute / 10)), (minute % 10) ]
Display.Show(val)
        Display.ShowDoublepoint((second % 2))
        time.sleep(0.25)
```

6. Results:

